PURPLE COMPANY

Project Title: Purple Company – Beauty Products E-Commerce Management System

1. Subqueries and Nested Queries

A subquery is a query written inside another SQL query.

```
SQL> SELECT Order_ID, Total_Amount
  2  FROM
  3  (
  4      SELECT Order_ID,
  5             SUM(Quantity * Unit_Price) AS Total_Amount
  6      FROM PurpleOrderItem
  7      GROUP BY Order_ID
  8  )
  9  WHERE Total_Amount >
 10  (
 11      SELECT AVG(Total_Amount)
 12      FROM
 13      (
 14          SELECT SUM(Quantity * Unit_Price) AS Total_Amount
 15          FROM PurpleOrderItem
 16          GROUP BY Order_ID
 17      )
 18  );

  ORDER_ID TOTAL_AMOUNT
---------- ------------
         1         6098
```

2. Find Products Above Average Price

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM PurpleProduct
  3  WHERE Price >
  4  (
  5      SELECT AVG(Price)
  6      FROM PurpleProduct
  7  );

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
         2
Liquid Foundation
       799

         4
Glow Blush
       599

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
```

Find Products Below Average Price

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM PurpleProduct
  3  WHERE Price <
  4  (
  5      SELECT AVG(Price)
  6      FROM PurpleProduct
  7  );

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
         1
Matte Lipstick
       499

         3
Waterproof Eyeliner
       349

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

         5
Volume Mascara
       449
```

3. Identify Customers with Maximum Purchases

Step 1: Calculate Customer-wise Purchase

```
SQL> SELECT O.Customer_ID,
  2         SUM(I.Quantity * I.Unit_Price) AS Total_Purchase
  3  FROM PurpleOrder O
  4  JOIN PurpleOrderItem I
  5  ON O.Order_ID = I.Order_ID
  6  GROUP BY O.Customer_ID;

CUSTOMER_ID TOTAL_PURCHASE
----------- --------------
          1           6098
          2           3000
          3           2400
          4           1800
```

Step 2: Find Maximum Purchase

```
SQL> SELECT MAX(Total_Purchase)
  2  FROM
  3  (
  4       SELECT SUM(I.Quantity * I.Unit_Price) AS Total_Purchase
  5       FROM PurpleOrder O
  6       JOIN PurpleOrderItem I
  7       ON O.Order_ID = I.Order_ID
  8       GROUP BY O.Customer_ID
  9  );

MAX(TOTAL_PURCHASE)
-------------------
               6098
```

3 Display Customer Details with Maximum Purchase

```
SQL> SELECT C.Customer_ID,
  2         C.Customer_Name,
  3         SUM(I.Quantity * I.Unit_Price) AS Total_Purchase
  4  FROM PurpleCustomer C
  5  JOIN PurpleOrder O
  6  ON C.Customer_ID = O.Customer_ID
  7  JOIN PurpleOrderItem I
  8  ON O.Order_ID = I.Order_ID
  9  GROUP BY C.Customer_ID, C.Customer_Name
 10  HAVING SUM(I.Quantity * I.Unit_Price) =
 11  (
 12      SELECT MAX(Total_Purchase)
 13      FROM
 14      (
 15          SELECT SUM(I.Quantity * I.Unit_Price) AS Total_Purchase
 16          FROM PurpleOrder O
 17          JOIN PurpleOrderItem I
 18          ON O.Order_ID = I.Order_ID
 19          GROUP BY O.Customer_ID
 20      )
 21  );

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_PURCHASE
--------------
         1
Arun
         6098
```

4. Complex Business Queries

4.1 Customers Who Placed Orders Above Average Order Value

```
SQL> SELECT DISTINCT C.Customer_ID,
  2         C.Customer_Name
  3  FROM PurpleCustomer C
  4  JOIN PurpleOrder O
  5  ON C.Customer_ID = O.Customer_ID
  6  JOIN PurpleOrderItem I
  7  ON O.Order_ID = I.Order_ID
  8  GROUP BY C.Customer_ID, C.Customer_Name, O.Order_ID
  9  HAVING SUM(I.Quantity * I.Unit_Price) >
 10  (
 11      SELECT AVG(Total_Amount)
 12      FROM
 13      (
 14          SELECT SUM(Quantity * Unit_Price) AS Total_Amount
 15          FROM PurpleOrderItem
 16          GROUP BY Order_ID
 17      )
 18  );

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
          1
Arun
```

5. Find Products Whose Sales Are Above Average

```
SQL> SELECT P.Product_ID,
  2         P.Product_Name,
  3         SUM(I.Quantity * I.Unit_Price) AS Total_Sales
  4  FROM PurpleProduct P
  5  JOIN PurpleOrderItem I
  6  ON P.Product_ID = I.Product_ID
  7  GROUP BY P.Product_ID, P.Product_Name
  8  HAVING SUM(I.Quantity * I.Unit_Price) >
  9  (
 10      SELECT AVG(Total_Sales)
 11      FROM
 12      (
 13          SELECT SUM(Quantity * Unit_Price) AS Total_Sales
 14          FROM PurpleOrderItem
 15          GROUP BY Product_ID
 16      )
 17  );

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_SALES
-----------
         1
Matte Lipstick
       4598

         3
Waterproof Eyeliner
       3000

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_SALES
-----------
```

6. Customer Purchase Report

```
SQL> SELECT C.Customer_ID,
  2         C.Customer_Name,
  3         COUNT(DISTINCT O.Order_ID) AS Total_Orders,
  4         NVL(SUM(I.Quantity * I.Unit_Price), 0) AS Total_Purchase,
  5         NVL(SUM(I.Quantity * I.Unit_Price), 0) /
  6         NULLIF(COUNT(DISTINCT O.Order_ID), 0) AS Average_Order_Value
  7  FROM PurpleCustomer C
  8  LEFT JOIN PurpleOrder O
  9  ON C.Customer_ID = O.Customer_ID
 10  LEFT JOIN PurpleOrderItem I
 11  ON O.Order_ID = I.Order_ID
 12  GROUP BY C.Customer_ID, C.Customer_Name
 13  ORDER BY Total_Purchase DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_ORDERS TOTAL_PURCHASE AVERAGE_ORDER_VALUE
------------ -------------- -------------------
           1
Arun
            1           6098                6098

           2
Kavin
            1           3000                3000

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_ORDERS TOTAL_PURCHASE AVERAGE_ORDER_VALUE
------------ -------------- -------------------

           3
Priya
            1           2400                2400
```

```
Anu

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_ORDERS TOTAL_PURCHASE AVERAGE_ORDER_VALUE
------------ -------------- -------------------
            1           1800                1800

           5
Riya
            1              0                   0
```

7. Advanced Product Sales Report

```
SQL> SELECT P.Product_ID,
  2         P.Product_Name,
  3         COUNT(DISTINCT I.Order_ID) AS Total_Orders,
  4         SUM(I.Quantity) AS Total_Quantity,
  5         SUM(I.Quantity * I.Unit_Price) AS Total_Sales,
  6         AVG(I.Unit_Price) AS Average_Price
  7  FROM PurpleProduct P
  8  JOIN PurpleOrderItem I
  9  ON P.Product_ID = I.Product_ID
 10  GROUP BY P.Product_ID, P.Product_Name
 11  ORDER BY Total_Sales DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_ORDERS TOTAL_QUANTITY TOTAL_SALES AVERAGE_PRICE
------------ -------------- ----------- -------------
         1
Matte Lipstick
           1              2        4598          2299

         3
Waterproof Eyeliner
           1              3        3000          1000

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_ORDERS TOTAL_QUANTITY TOTAL_SALES AVERAGE_PRICE
------------ -------------- ----------- -------------

         4
Glow Blush
           1              2        2400          1200

         5
Volume Mascara
```

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_ORDERS TOTAL_QUANTITY TOTAL_SALES AVERAGE_PRICE
------------ -------------- ----------- -------------
           1              1        1800          1800

         2
Liquid Foundation
           1              1        1500          1500
```

8. Category-wise Advanced Sales Report

```
SQL> SELECT C.Category_ID,
  2         C.Category_Name,
  3         SUM(I.Quantity) AS Total_Quantity,
  4         SUM(I.Quantity * I.Unit_Price) AS Total_Sales
  5  FROM PurpleCategory C
  6  JOIN PurpleProduct P
  7  ON C.Category_ID = P.Category_ID
  8  JOIN PurpleOrderItem I
  9  ON P.Product_ID = I.Product_ID
 10  GROUP BY C.Category_ID, C.Category_Name
 11  ORDER BY Total_Sales DESC;

CATEGORY_ID
-----------
CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY TOTAL_SALES
-------------- -----------
          1
Lipstick
             2        4598

          3
Eyeliner
             3        3000

CATEGORY_ID
-----------
CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY TOTAL_SALES
-------------- -----------

          4
Blush
             2        2400

          5
Mascara
```

```
CATEGORY_ID
-----------
CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY TOTAL_SALES
-------------- -----------
             1        1800

          2
Foundation
             1        1500
```